



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

AZURE BICEP PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A DevOps

TOTAL: 10 QUESTIONS

Q1 What problem does Azure Bicep solve in DevOps or infrastructure?

Threat Model

Q6 How does Azure Bicep integrate with CI/CD pipelines?

Observability

Q2 What is Azure Bicep's core architecture?

Architecture

Q7 How do you debug a failed Azure Bicep deployment?

Lifecycle

Q3 How does Azure Bicep define infrastructure or configuration as code?

Configuration

Q8 What are security best practices for Azure Bicep?

Compliance

Q4 How does Azure Bicep ensure idempotency?

Hardening

Q9 How does Azure Bicep scale for large environments?

Testing

Q5 How do you handle secrets in Azure Bicep?

SSO / IdP

Q10 What are common mistakes teams make when adopting Azure Bicep?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Azure Bicep addresses a specific concern

Mitigation

Azure Bicep addresses a specific concern—provisioning, configuration management, orchestration, or CI/CD—by automating manual steps that are error-prone, slow, or not reproducible when done by hand.

A6 CI/CD pipelines call Azure Bicep in

Observability

CI/CD pipelines call Azure Bicep in automated steps: plan on a pull request to preview changes and apply on merge to main. Approval gates can require a human to review the plan before changes go to production.

A2 Azure Bicep has a control plane

Primitives

Azure Bicep has a control plane that stores desired state and a data plane (agents or push-based SSH) that enforces it. Understanding this distinction clarifies where configuration is evaluated and where changes are applied.

A7 Check Azure Bicep's logs for the

Automation

Check Azure Bicep's logs for the specific error message and the resource that failed. For infrastructure tools, re-run with increased verbosity (-v, --debug). Review the state file or event history to understand what changed before the failure.

A3 Azure Bicep uses declarative files (YAML,

Hardening

Azure Bicep uses declarative files (YAML, HCL, or a DSL) to express desired state. A plan/diff phase shows what will change before applying. Version-controlling these files enables peer review and rollback.

A8 Rotate credentials used by Azure Bicep,

Governance

Rotate credentials used by Azure Bicep, restrict network access to its control plane, enable audit logging of all operations, and use least-privilege service accounts. Never run Azure Bicep with admin credentials in production.

A4 Azure Bicep operations are designed to

Gotchas

Azure Bicep operations are designed to produce the same result when run multiple times. Applying the same config twice converges to the desired state rather than creating duplicate resources. This makes automation safe to re-run.

A9 Large Azure Bicep deployments use

Assessment

Large Azure Bicep deployments use workspaces or namespaces to isolate environments, remote state backends for team collaboration, and modular code to avoid a single monolithic config that is slow to plan/apply.

A5 Secrets should never be stored in

Federation

Secrets should never be stored in plain-text in Azure Bicep config files. Use a secrets backend (Vault, AWS Secrets Manager) and inject values at runtime via environment variables or encrypted vaults supported by the tool.

A10 Common pitfalls include storing secrets in

Optimization

Common pitfalls include storing secrets in plain-text config, skipping the plan step before apply, not reviewing state drift, and not having a rollback strategy when a deployment fails partway through.